SYSTEMS THINKING & DESIGN (PHASE 3-4 Submission)

WHERE THE SYSTEM COULD BE CHANGED

Leverage points, intervention opportunities and design principles for shared resources in SERC and Precog.

TEAM: THE SYSTEM ERROR

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Phase 2 found a readability problem sitting on top of a capacity problem

What Phase 2 taught us

The same pattern kept appearing: a desk stays empty while still treated as someone’s, a GPU slot is held so someone “might run the experiment some day”, and a junior sees a free seat and still doesn’t sit. Helpfulness is high and faculty act quickly when a problem reaches them. What fails is the path between an available resource and the person who needs it.

Why scarcity alone is incomplete

Peak-time shortage is real, and more capacity would help at the margin. But new desks or GPUs would not change how claims are made, how status is read or who feels able to ask. Under the same rules, added capacity would be claimed the same way.

Why Phase 3 looks at structure

Defensive reserving, silent yielding and blanket restriction are produced by rules, information flows and roles, not by selfish individuals, an assumption Phase 2 discarded. So Phase 3 asks where those structures could change, and who could change them.

Why several leverage points, not one fix

Our resources behave differently: a running GPU job, a desk with a monitor on it, a fridge and a coffee machine do not need the same rule. Several causal links are still hypotheses. Keeping more than one leverage point open lets Phase 4 test them against each other instead of committing early.

Five mechanisms produce the outcomes we observed

MECHANISM	WHAT WE OBSERVED	STRUCTURE BEHIND IT	RESULTING BEHAVIOUR	LOOP / ROOT CAUSE
M1 **Claims persist**	Names and belongings mark desks; holders keep return priority; GPU slots are held weeks ahead of use.	No shared release or expiry condition for desks or idle reservations.	Empty ≠ available. Users reserve earlier and longer, most of all near deadlines.	R1 , R3 , R4
M2 **Status is unreadable**	SERC uses a tracker, Precog uses Slack, desks use names. Users check, then still message the holder.	No common view of who holds what, since when, or how to reach them.	Every check is a person-to-person query; an idle slot looks like an active one.	R2, RC3
M3 **Access runs on asking**	Juniors hesitate at desks they were never refused; GPU priority is won by persuasion on Slack.	Permission is informal, held by incumbents and seniors; norms travel by word of mouth.	Offered permission ≠ felt permission. Access follows confidence and network.	R2 , RC2, RC7
M4 **Problems surface late; correction is blunt**	Complaints stay in WhatsApp; notices appear after breakdowns; coffee access was withdrawn from all students.	No channel returns small, recurring issues to someone who can act; restriction is the only correction.	Authority stays under-informed, then responds broadly; the shared benefit is lost.	B1, RC9, RC10
M5 **Allocation follows history and funding**	Desk arrangement between labs predates current headcount; compute access follows the professor - a “hard rule, no exceptions”.	No standing process to revisit allocation; boundaries are set by who paid.	Idle capacity in one group while another is short.	R3, RC1, RC5, RC8

BENEATH ALL FIVE (MENTAL MODELS)

“Resources bought by a professor belong to that group” · “A desk someone regularly uses is effectively theirs” · “Small conflicts should be solved among students”. Some of these beliefs also do useful work - respect for others’ setups and a preference for peaceful dialogue — so the aim is to redirect them, not remove them.

LEVERAGE POINT MAP

HOW TO READ
- ⚠ Current problem
- ↻ Root cause / loop
- Leverage point
- Intervention
- ✓ Desired effect
- Deep structural
- enables (direct dependency)
- shapes (deep influence)
- R2 = causal loop targeted

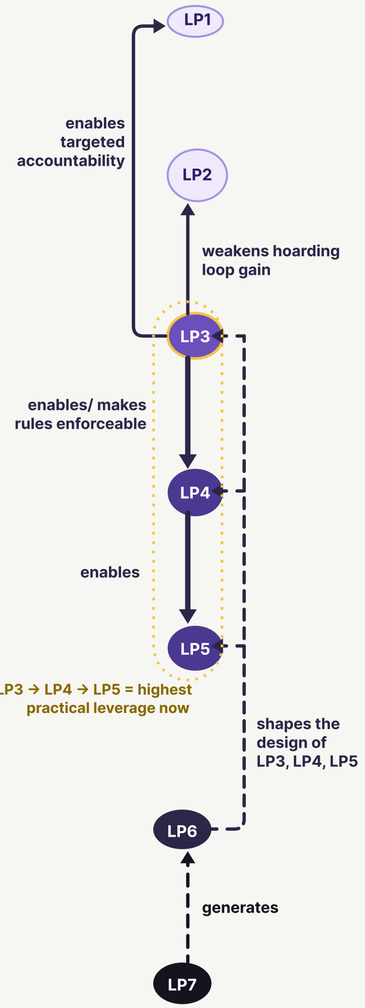
SHALLOW
Fast, Easy, Lower structural leverage

- EVENTS
- PATTERNS
- STRUCTURES LP3 - LP5
- GOALS
- MENTAL MODELS

DEEP
Harder, Slower, Higher structural leverage

⚠ CURRENT BEHAVIOUR	↻ ROOT CAUSE / LOOP	LEVERAGE POINT	INTERVENTION DIRECTION	✓ EXPECTED SYSTEM EFFECT
One person's misuse = everyone gets restricted. e.g. coffee machine access withdrawn; fridge policy tightened for all after isolated incidents.	Collective punishment as the default correction mechanism Targets B1	LP1 SHALLOW **Correction Targeting** LP3 enables targeted accountability here	Target correction at the responsible individual, not the entire group.	More proportional accountability · less resentment · unnecessary restriction
A scarce resource can be held idle at effectively zero cost. Others can't see if it's idle, how long it's been held, or if it's still needed	Idle holding carries no visible cost Targets R1 R3 R4	LP2 SHALLOW → MEDIUM **Loop Gain on Hoarding** Weakened by LP3 (idle-holding becomes visible)	Expose idle-holding: make prolonged unused holding visible so it carries a social / reputational cost.	Reduces the momentum of scarcity-driven hoarding
SERC tracker few use consistently, Precog runs on ad-hoc Slack DMs, no cross-lab visibility, every check = person-to-person effort	Informality → information asymmetry. Free resources are socially unavailable. Targets R2	LP3 CORE · MEDIUM **Information Flow** ★ HIGHEST-LEVERAGE BUILDABLE INTERVENTION Enables LP1, LP2, LP4	One shared, low-friction visibility layer: resource, holder, active/ idle, time held, expected release, contact	Availability becomes readable. Unlocks LP2, LP1 and LP4.
Rules mostly unwritten: seniority entitlement, no time-boxed reservations, no no-show consequence, no light escalation path	Reliance on unwritten, unenforced courtesy norms	LP4 MEDIUM → DEEP **Rules of Engagement** Depends on LP3: can't enforce what no one can see	Explicit lightweight rules: reservation expiry, idle-release, absence-triggered desk release, documented escalation path	Behaviour moves from psychological negotiation to visible system rules
Allocation is historically frozen: the SERC-Precog desk split doesn't adapt to changing headcount.	Allocation ossification	LP5 DEEP STRUCTURAL **Self-Organization Capacity** Enabled by LP4	Standing mechanisms that let the system adapt, e.g. semesterly reallocation review on current headcount + usage	System adjusts as demand changes, instead of one-off negotiation
▼ DEEP STRUCTURAL CONDITIONS : they decide whether LP3–LP5 persist, not buildable product features				
Implicit goal: "Protect what my lab/group paid for." ownership-first behaviour, blanket cross-lab denial.	Generated by the LP7 mental model	LP6 DEEP **System Goal** Shapes the design of LP3, LP4, LP5	Shift the operating goal to UTILIZATION WITH ATTRIBUTION: attribution remains; exclusivity reduces where capacity sits idle. Not "remove ownership".	Rules, incentives, info and processes optimise utilization, not territorial protection
Beliefs: "Resources bought by a group belong to that group." "Seniority gives stronger entitlement."	Deepest condition: shapes whether LP3–LP6 persist	LP7 DEEPEST **Mental Model** Generates LP6	Design interventions that preserve attribution without turning attribution into exclusivity.	Decides whether the system holds its gains or regresses

DEPENDENCY CHAIN



Rules, roles, incentives, information and processes

Rules

- Idle GPU reservations expire after a grace window; running jobs untouched (a 6-hour interactive cap already exists)
- Desks carry an expected-return date; long absence makes them reclaimable, not forfeited
- Consequences attach to the individual, not the group
- A request path for idle cross-lab compute, decided by that resource’s POC

LP2, LP4, LP1, LP6

Roles

- A named contact per resource type, building on the existing GPU manager and PhD POCs
- A cross-lab contact for shared-space and reallocation decisions
- Escalation ladder: peer → resource POC → faculty
- POC briefs new joiners on local norms

LP4, LP5

Incentives

- Idle holding is visible, so it carries a small social cost
- Early release is acknowledged.
- Deadline priority follows stated criteria, not who argues best

LP2

Information flows

- One readable view of status, time held and contact across both labs
- A record of reserved vs. actually used time, so patterns stop being anecdotal

LP3, LP1

Processes

- Request → time-boxed use → confirmed release as one sequence
- Semesterly desk review against headcount
- Staged conflict handling matching the ladder

LP4, LP5

Why each intervention exists: from mechanism to risk

A makes B checkable — a rule no one can see is just another unwritten norm

	A Make status readable	B Let claims end, per resource	C Give small needs a route	D Correct misuse, keep benefit	E Let allocation follow use
CURRENT SYSTEM MECHANISM	M2: Status sits in separate places- SERC tracker, Precog Slack, names on desks. An idle slot looks like an active one; every check is a message.	M1: Once made, a claim persists: names and belongings on desks, return priority, GPU slots held for “some day”.	M3: Access runs on asking. Juniors hesitate at desks they were never refused; priority goes to the most persuasive.	M4: Misuse surfaces as a breakdown; the response is restriction for everyone - coffee access withdrawn from all students.	M5 · Desk arrangement predates current headcount; compute access follows the professor. Idle capacity sits beside shortage.
LEVERAGE POINT	**LP3 Information flow + LP2 Loop gain**	**LP4 Rules + LP2 Loop gain**	**LP4 Roles and decision rights**	**LP1 Balancing-loop targeting**	**LP5 Self-organisation + LP6 Goal**
INTERVENTION OPPORTUNITY	Make holder, expected return and contact readable across both labs - building on the tracker and Slack, not replacing them.	Release conditions that fit each resource: expected-return dates for desks, idle-reservation limits for GPUs - running jobs untouched.	Name who to ask per resource, add a step between peers and faculty, and tell newcomers the local norms.	Attach notices and consequences to the individual and keep shared access - building on the Saturday fridge rule.	A semesterly desk review against headcount; a request path for idle cross-lab compute, decided by its own POC.
LOOP / ROOT CAUSE AFFECTED	R2, R1 | RC3 visibility	R1, R3, R4 | no release rule	R2, B2 | RC2, RC9	B1 | RC10 trust vs control	R3 | RC1, RC5,RC8
EXPECTED SYSTEM EFFECT	Fewer manual checks; idle holding gets noticed; free capacity reads as free.	Claims track actual use; less defensive reserving before deadlines.	Asking becomes cheaper and less personal; unmet need reaches someone who can act.	Misuse still controlled; shared amenities stay open to students.	Allocation adapts before crises; idle capacity used with attribution.
POSSIBLE UNINTENDED CONSEQUENCE	*A register makes indefinite claims look official; exposure turns into shaming; a third parallel system.*	*Interrupts long experiments or legitimate absence; token check-ins to keep a hold.*	*The contact becomes a new bottleneck or unpaid work; courtesy replaced by procedure.*	*Identifying users feels like surveillance; more follow-up work for cleaning staff.*	*Sponsors see loss of control; review becomes symbolic; friction between groups.*
WHO CAN ACT	GPU manager, POCs, students	Faculty + POCs set rules, holders	POCs, seniors, faculty	Faculty, admin, supervisor & cleaning staff	Faculty of both labs · sponsors · admin

UNDERNEATH EVERY BRANCH: LP6 SYSTEM GOAL + LP7 MENTAL MODELS

Goal (inferred): “protect what my group funded” → “use idle capacity, with attribution”. Beliefs: “bought by a group, belongs to the group” · “a desk I use regularly is mine” · “small issues shouldn’t reach faculty”. If these stay unnamed, the changes in A–E can drift back into ownership behaviour.

From Leverage Point to Intervention

LEVERAGE PT	SYSTEM DIMENSION	⚠ CURRENT STATE	PROPOSED CHANGE	↻ LOOP / ROOT CAUSE TARGETED
LP4	**RULES** 4 changes	GPU reservations have no time limit.	Reservation expiry + grace window + auto-release.	R1 scarcity → hoarding
LP4		Desk becomes effectively permanent once claimed.	No check-in for a defined period → desk becomes reclaimable.	Allocation ossification
LP1		One person's misuse leads to blanket restriction.	Consequences attach to the identified individual.	B1 Collective punishment
LP6		Cross-lab compute access is assumed impossible.	Lightweight documented request path for idle capacity; the resource POC decides.	R3 scarcity → ownership
LP5	**ROLES** 4 changes	One faculty member is the default escalation point.	Student resource coordinator per resource type (GPU, desks, fridge).	Single escalation point
LP5		Cross-lab issues handled through one-off negotiation.	Standing SERC–Precog liaison role.	One-off negotiation
LP4		Only peer resolution or faculty escalation.	Peer → resource POC → faculty.	Reluctance to escalate
LP4		New students learn norms informally.	Resource POC responsible for lightweight onboarding.	Unwritten courtesy norms
LP2	**INCENTIVES** 3 changes	Idle holding has zero visible cost.	Expose prolonged idle holding → creates social / reputational cost.	R1 R3
LP2		No recognition for early release.	Lightweight public acknowledgement for releasing / sharing capacity → reinforces cooperation.	Scarcity-driven hoarding
LP4		Deadline priority is negotiated informally.	Transparent deadline-aware priority → depends less on seniority or persuasion.	R4 Seniority entitlement
LP3	**INFORMATION FLOWS** 3 changes	SERC tracker and Precog Slack are disconnected.	One cross-lab visibility layer: status · holder · time held · contact.	R2
LP3		Checking if a GPU is genuinely active needs manual verification.	Automatically surface active / idle state and holder.	R2 Information asymmetry
LP3		No record of reserved time vs actual usage.	Log the reservation-to-usage gap → evidence for targeted correction + hoarding reduction.	R1 Collective punishment
LP5	**PROCESSES** 3 changes	Desk / locker allocations persist historically.	Semesterly reallocation review based on current headcount and usage.	Allocation ossification
LP4		Request, use and release are disconnected informal actions.	REQUEST → TIME-BOXED USE → CONFIRMED RELEASE as one continuous process.	R1 Informal negotiation
LP4/LP5		Conflict resolution is undefined.	PEER → RESOURCE POC → FACULTY as an explicit escalation process.	B2 Conflict

Design Principles

01 **Make availability visible before making it enforceable**

Rules only work when people can see whether they are being followed.

PREVENTS: Unwritten norms that no one can check (R2, M2)

02 **Correct the actor, not the population**

Consequences land on the person responsible, not on everyone.

PREVENTS: Collective restriction that removes the shared benefit (B1, M4)

03 **Let claims decay by default**

A hold needs an ongoing signal of use, not a one-time claim.

PREVENTS: Persistent claims and idle occupation (R1, R3, M1)

04 **Route unmet need to whoever can act on it**

A proportionate step between peers and faculty.

PREVENTS: Needs lost in WhatsApp or silent yielding (RC9, M3)

05 **Design for actual usage, not historical allocation**

Allocation updates as the population does.

PREVENTS: Desk arrangements frozen at an old split (RC8, M5)

06 **Preserve attribution while enabling sharing**

Credit and control stay visible; idle capacity can still flow.

PREVENTS: Ownership treated as exclusivity (RC1, RC5)

07 **Favour self-serve, lightweight mechanisms over added approval**

Reduce the friction of asking; don’t add another layer.

PREVENTS: New processes ignored, as escalation to faculty is now (M3)

Potential Intervention Points

1 Shared resource visibility board

▲ HIGH PRIORITY

DESCRIPTION

One cross-lab view showing current holder · time held · idle status · contact method.

Replaces the SERC tracker / Precog Slack split.

WHY A STRONG CANDIDATE

Directly targets unreadable availability. Additive. Low bureaucracy. Easy to test.

PRINCIPLES EMBODIED

P1 · Visibility first | P3 · Claims decay | P6 · Attribution + sharing

LEVERAGE POINT TARGETED

LP3

★ IN PROTOTYPE FOCUS → 04

2 Auto-expiring desk / locker tags

▲ HIGH PRIORITY

DESCRIPTION

Name + expected return date. If no return / check-in occurs, the resource becomes reclaimable.

WHY A STRONG CANDIDATE

Cheap to prototype. Directly addresses permanent claims.

PRINCIPLES EMBODIED

P3 · Claims decay | P5 · Actual usage

LEVERAGE POINT TARGETED

LP4 | LP5

★ IN PROTOTYPE FOCUS → 04

3 Individual-accountability policy for shared consumables

● MEDIUM PRIORITY

DESCRIPTION

Logging + individual notification before restriction / discard

WHY A STRONG CANDIDATE

Low effort. Reverses the documented collective-punishment failure.

PRINCIPLES EMBODIED

P2 · Target the actor

LEVERAGE POINT TARGETED

LP1

4 Rotating resource coordinator

▲ HIGH PRIORITY

DESCRIPTION

Student-held lightweight authority for each resource category.

WHY A STRONG CANDIDATE

Reduces reliance on a single faculty escalation point.

PRINCIPLES

P4 · Route the need | P7 · Self-serve

LEVERAGE POINT TARGETED

LP5

★ IN PROTOTYPE FOCUS → 04

5 Peer → coordinator → faculty escalation ladder

● MEDIUM PRIORITY

DESCRIPTION

Defined, proportionate conflict path.

PRINCIPLES

P4 · Route the need | P7 · Self-serve

LEVERAGE POINT TARGETED

LP4 | LP5

6 Semesterly reallocation review

○ LOW · CURRENT TERM | ◆ HIGH STRUCTURAL VALUE

DESCRIPTION

Periodic headcount- and usage-based resource allocation review.

WHY A STRONG CANDIDATE

Important structural fix, but slow to test.

PRINCIPLES EMBODIED

P5 · Actual usage

LEVERAGE POINT TARGETED

LP5

7 New-member onboarding checklist

● MEDIUM PRIORITY

DESCRIPTION

Short onboarding covering desk norms · shared-resource norms · fridge policy · escalation path.

WHY A STRONG CANDIDATE

Very cheap. Addresses accidental violations by new students.

PRINCIPLES

P1 · Visibility first | P7 · Self-serve

LEVERAGE POINT TARGETED

LP4

KEY

Design principles

- **P1** Visibility first: Make availability visible before making it enforceable.
- **P2** Target the actor: Correct the actor, not the population.
- **P3** Claims decay: Let claims decay by default.
- **P4** Route the need: Route unmet need to whoever can actually act on it.
- **P5** Actual usage: Design for actual usage, not historical allocation.
- **P6** Attribution + sharing: Preserve attribution while enabling sharing.
- **P7** Self-serve: Favor self-serve, lightweight mechanisms over added administrative approval.

Prototype priority

▲ HIGH PRIORITY | ● MEDIUM PRIORITY | ○ LOW · CURRENT TERM

Priority is about what we can build and observe this term. Depth (01) is about structural leverage. They are separate axes.

Alternative Interventions

INTERVENTION	DEPTH TYPE	MECHANISM · LEVERAGE POINT	[P] WHAT CHANGES	MAIN NEW BURDEN
1 · Claim Signals	Light · information flow	R2 informality → information asymmetry · LP3	Standard claim states on desks (In use / Away until / Free to borrow / Protected) and two fields on the existing GPU-ownership site (expected release, OK to pause?).	Holders update markers
2 · Default-Decay Claims	Moderate · rule	R1 desk spiral, hoarding · LP4 (+LP2)	Claims lapse unless used or renewed; continuity declared upfront with an end date; GPU reservations expire if unstarted.	Holders renew; someone confirms lapses
3 · Proportionate Care Protocol	Light · rule + process	B1 misuse → blanket restriction · LP1	Label → reminder → private correction → individual restriction; blanket restriction only with a written reason and review date.	Reminder-sender; faculty
4 · Resource Stewards + Escalation Ladder	Moderate · role + routing	B2 + broken feedback link · LP4 → LP5	Named student stewards per resource with limited written decision rights; peer → steward → faculty; one-line issue log.	Stewards (largest new load)
5 · Periodic Allocation & Issue Review	Deep · governance	Ossification, ownership boundary · LP5 (for LP6)	Short joint SERC–Precog review each term using the issue log: desk split vs headcount, recurring issues → rule changes, cross-lab idle-GPU path.	Faculty time

[I] Dependency chain: decay (2) is only enforceable if lapses are visible (1) and someone other than faculty confirms them (4); the review (5) needs the steward log (4). Concept 3 stands alone for amenities.

PROTOTYPE P1 · PHYSICAL SIGNAL

Desk claim card: the state of a claim, readable from the aisle

Extends the existing desk name-writing system. One A5 card in a clear sleeve at the desk edge; the holder ticks one state and writes a date.

DESK 27 · SERC BAY B HOLDER: _____

☐ **IN USE**
I work here most days.

☐ **AWAY UNTIL __ / __**
Planned absence, max 14 days without a continuity exception.

☐ **FREE TO BORROW TODAY**
Use it and leave it as you found it.

☐ **PROTECTED SETUP · ASK FIRST**
Ends __ / __ · steward initials ____ (needs an end date)

LAST HERE: __/__ __/__ __/__ __/__ __/__

Date older than 7 days → anyone may borrow for the day. Questions: desk steward (name on bay notice).

WHERE · WHO SEES IT

Desk edge facing the aisle; seen by anyone walking the bay, especially newcomers.

INFORMATION IT COMMUNICATES

Claim state, planned return, whether borrowing is welcome, and when a protected setup ends. Not what the person is doing.

BEHAVIOUR IT CHANGES

Replaces the jacket-on-chair signal; lets newcomers use the ask-first norm with confidence; makes a stale claim visible.

HOW IT CHANGES STATE

Holder ticks a state and dates "last here" at least weekly. Protected needs an end date and a steward's initials.

WHO MAINTAINS IT

Holder. The desk steward re-sleeves missing cards on a weekly walk; nobody logs individuals.

IF IT GOES OUT OF DATE

A date older than 7 days reads as "borrowable today"; older than 21 days triggers a steward message (P3 policy).

FAILURE LOOKS LIKE

Cards left blank; every card ticked Protected; borrowers still afraid to sit. Measured on the weekly walk.

PROTOTYPE P2 · WRITTEN RULE

Shared Desk, Locker & GPU Claims: how it works

Can I use a desk?

Yes if its card says FREE TO BORROW, or its "last here" date is older than 7 days. Borrow for the day and leave it as found. For a desk of your own, ask the desk steward to add you to the queue.

Do I need permission?

Not to borrow a free or stale desk. Asking first is still the courteous default. A PROTECTED desk is never borrowed without the holder saying yes.

How long can I keep a claim?

As long as you use it. Date your card at least weekly. Away up to 14 days: tick AWAY UNTIL. Longer: request a continuity exception.

When is a claim released?

Card date older than 21 days → the steward messages you. No reply in 5 working days → the desk returns to the queue. Lockers: renew in the first 2 weeks of term.

GPUs?

Existing quota tiers and the 6-h interactive cap stay. Every reservation shows an expected release. A reservation not started within 2 h lapses. Running jobs are never stopped by this rule.

What if I need an exception?

Continuity exception with an end date: up to 4 weeks, the steward approves; longer, faculty approves. Deadline priority for GPUs needs a dated deadline within 72 h.

Who do I contact?

Your peer first. Then the steward for that resource (names on the bay notice and GPU page). Then faculty, if the steward's decision is contested.

WEAKNESS OF THE CURRENT NORM

Claims last until contested, and contesting costs juniors socially [E]. Courtesy works for insiders only.

ENABLES

Same-day borrowing without a broker; recovery of dead claims; open protection of real continuity.

DISCOURAGES

Holding without use; unstarted GPU reservations; persuasion-based priority.

FLEXIBILITY KEPT

Lending, asking first, and steward or faculty judgment on every exception.

IF THE RULE IS NOT FOLLOWED

Signal (card or page) → reminder from steward → conversation → faculty. Never automatic removal of belongings or a running job.

OUT OF SCOPE

Faculty-assigned desks, meeting rooms, ownership of purchased equipment, the lab desk split (termly review).

PROTOTYPE P3 · RULE + PROCESS (POSTED AT THE FRIDGE)

Proportionate care: correct the actor, not the population

1 · SIGNAL

Label + date every item (sticker station beside the fridge). The Saturday clear stays as the weekly reset.

→

2 · REMINDER

Thursday: the amenities steward posts a list of unlabelled or expired shelves. No names.

→

3 · PRIVATE WORD

Same labelled owner twice in a term → a private word from the steward. Not public naming.

→

4 · INDIVIDUAL LIMIT

Third time → the steward asks faculty for a time-limited individual restriction (e.g. 2 weeks).

→

LAST RESORT · BLANKET

Faculty only, with a written reason and a review date within 4 weeks, posted here.

CLEANING & SERVICE STAFF

May send one photo of a recurring mess to the amenities steward. They report; they never confront or decide [E: staff execute, never decide].

COFFEE MACHINE [H]

A proposal to faculty, not a rule change: a conditional student trial with named users and a care rota, reverted if damage recurs.

WHAT IT TESTS

Whether misuse can be attributed through labels alone (no cameras, no logs of people), and whether the step order feels fair.

WHAT IT CANNOT TEST

Anonymous misuse; faculty willingness to relax existing restrictions.

PROTOTYPE P4 · GOVERNANCE

Who can decide what: the decision-rights matrix

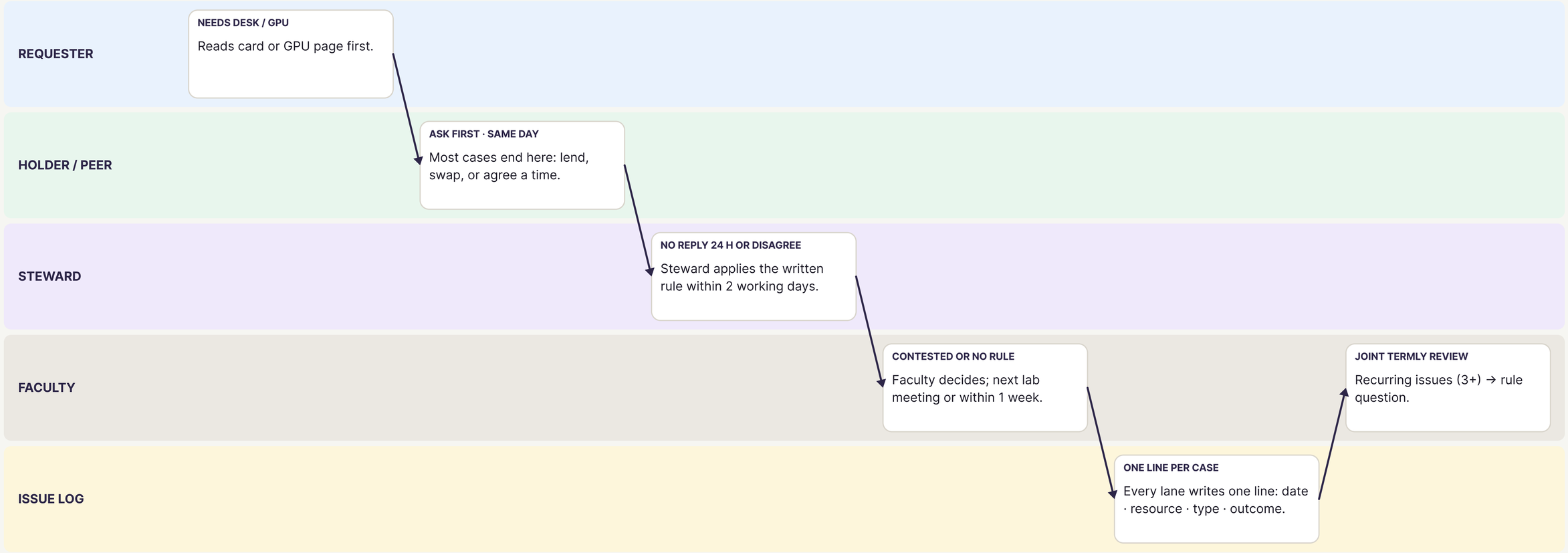
D = decides · C = consulted · I = informed. Stewards decide only what a written rule covers; everything else stays with faculty. Stewards serve in cross-lab pairs, one term each, with a time cap of 1 h/week.

DECISION	HOLDER / PEER	DESK · AMENITY STEWARD	GPU STEWARD (existing POC)	LAB FACULTY	JOINT TERMLY REVIEW	LAB ADMIN
Lend your own desk or GPU slot for a day	D	—	—	—	—	—
Borrow a desk marked free, or stale 7+ days	D (borrower)	I	—	—	—	—
Return a desk to the queue (21 days + 5 no reply)	C (holder)	D	—	I	I	—
Continuity exception up to 4 weeks	C	D (desks)	D (GPUs)	I	—	—
Continuity exception over 4 weeks	C	C	C	D	I	—
GPU deadline priority (dated ≤72 h)	C (consent)	—	D	I	—	—
Individual amenity restriction	C	C	—	D	I	—
Blanket restriction of an amenity	—	C	—	D (written, dated)	I	I
Cross-lab request for idle GPU capacity	C (requester)	—	D (owning lab)	I	I	—
Permanent desk reassignment · lab desk split	—	C	—	C	D	I
Report and fix a fault (dead desk, fridge)	I (reporter)	Log + route	—	I	I	D (fix)
Change any number in the policy	C	C	C	C	D	—

PROTOTYPE P5 · PROCESS

Escalation ladder: peer → steward → faculty

Shows who acts, who decides, who carries the load and where delay sits. Most cases should end in the first lane, as they do today.



WHERE DELAY SITS

Steward response (≤2 working days) and faculty (≤1 week). Today the delay is unbounded and often invisible.

WHERE COORDINATION IS REDUCED

No senior broker needed; one named contact per resource instead of DMs and WhatsApp.

WHERE HUMAN JUDGMENT REMAINS

Every exception, every contested case and every restriction is decided by a person, never by the rule alone.

WHO CARRIES THE LOAD

Stewards (capped at 1 h/week) and a little extra for faculty. Juniors carry less social risk than today.

DEVELOP SCENARIOS · SCENARIO MODEL

Eight scenarios drawn from what our research actually found

Each scenario is evaluated analytically against the prototypes. Outcomes on the next slides are reasoned predictions from existing evidence, not observed results.

SCENARIO	CATEGORY	TRIGGER	PROTOTYPE USED	WHAT IT EXAMINES
S1 · Newcomer needs a desk	Newcomer · normal use	First week on the floor; no desk assigned.	P1 card · P2 policy · onboarding	Readability without insider knowledge
S2 · Deadline week on the GPUs	Peak demand	Conference deadline in 48 h; every A40 slot reserved.	P2 policy	Rules under stress; steward load; false urgency
S3 · Long run + planned travel	Continuity	Leaving for a 2-week conference mid-run.	P1 Protected state · P2 exception	Continuity protection; renewal abuse
S4 · Two legitimate users, one resource	Conflict	Both want the same borrowable window desk, or both have dated deadlines.	P5 swimlane · P4 matrix	Steward legitimacy; decision by rule
S5 · Holder renews without using	Non-compliance · gaming	Card renewed as AWAY UNTIL for 6 weeks; desk visibly unused.	P2 rules	Ritual renewal; steward social cost
S6 · Holder unreachable	Resource owner absent	Reservation made, job not started, holder offline.	P2 GPU rule	Default decay without contacting anyone
S7 · Precog student needs idle SERC GPU	Cross-lab request	Overnight run needed; a SERC A40 shows idle.	P4 matrix · mock review	Ownership boundary; LP6 goal in practice
S8 · Dead desk and fridge mess	Maintenance failure	A desk with no light or fan; unlabelled food recurring.	P3 care card · P5 swimlane	Routing; feedback loop; staff role

SCENARIO TEST · 1 OF 3

Current system vs intervention, scenario by scenario

Predicted from existing evidence and the prototypes. [E] marks the current-system behaviour our research documented.

SCENARIO · ACTORS	CURRENT SYSTEM	WITH INTERVENTION	DECISION POINTS	LIKELY FAILURE MODE	EXPECTED OUTCOME
S1 · Newcomer needs a desk **New MS student, desk steward, holders in the bay**	Walks the bay reading names and jackets; asks a senior who knows who is away; may sit at a claimed desk and be moved on [E].	Reads cards: sits at a FREE or 7-day-stale desk; asks the desk steward to join the queue; gets a 10-minute norms briefing.	Is the card stale? Is the desk Protected? Is the queue fair to earlier arrivals?	Cards not updated, so ambiguity remains; or every card ticked Protected.	Seated the same day without a broker; on the queue for a desk of their own.
S2 · Deadline week on the GPUs **Two students with deadlines, holders, GPU steward**	Urgency persuasion through DMs; seniority and ties decide; idle-but-held slots stay held [E].	Unstarted reservations lapse after 2 h; the page shows expected release; deadline requests go to the GPU steward, who asks "OK to pause" holders first.	Does the deadline qualify (dated, ≤72 h)? Who pauses, and with whose consent?	Everyone declares a deadline; the steward is swamped; holders stop marking "OK to pause".	Idle-held capacity recovered; priority depends less on persuasion.
S3 · Long run + planned travel **PhD student with a 10-day training run and a desk setup**	Jacket and name hold the desk; nobody knows if the run is live; others avoid or contest it.	Declares a continuity exception with an end date on the card and GPU page; the steward approves (≤4 weeks).	Exception length; renewal after return; what counts as "use".	Exceptions renewed back-to-back become a permanent claim under a new name.	Setup protected openly; others know exactly when it frees.

SCENARIO TEST · 2 OF 3

Current system vs intervention, scenario by scenario

Predicted from existing evidence and the prototypes. [E] marks the current-system behaviour our research documented.

SCENARIO · ACTORS	CURRENT SYSTEM	WITH INTERVENTION	DECISION POINTS	LIKELY FAILURE MODE	EXPECTED OUTCOME
S4 · Two legitimate users, one resource **Two students, desk or GPU steward, faculty**	First-come or seniority; unresolved grievances go to WhatsApp and never reach faculty [E].	Borrowing is first-come for the day; permanent desks go by queue order; the steward decides by rule within 2 days; contested cases go to faculty; logged.	Which rule applies? Is it contested? Who informs whom?	A junior steward avoids ruling against a senior; the case bounces to faculty.	Decided by a stated rule and recorded; repeated clashes become visible.
S5 · Holder renews without using **Holder, desk steward**	Nobody challenges; the desk stays dead; the R1 desk spiral tightens [E].	A second consecutive renewal without dated use → steward conversation → continuity exception or return to the queue after notice.	What counts as use? When does a conversation become a decision?	The steward avoids the conversation; the holder removes the card to "go dark".	Persistent claims become discussable, and are handled privately, not publicly.
S6 · Holder unreachable **GPU holder at a conference, waiting student**	Others wait and DM repeatedly; job ownership is invisible [E].	The reservation lapses 2 h after its start time; a running job is untouched; the holder sees "lapsed" when back.	Was the job legitimately delayed? Is the holder's grace fair?	A false lapse when a job was about to start; the holder starts dummy jobs to hold slots.	Capacity returns without a message chain.

SCENARIO TEST · 3 OF 3

Current system vs intervention, scenario by scenario

Predicted from existing evidence and the prototypes. [E] marks the current-system behaviour our research documented.

SCENARIO · ACTORS	CURRENT SYSTEM	WITH INTERVENTION	DECISION POINTS	LIKELY FAILURE MODE	EXPECTED OUTCOME
S7 · Precog student needs idle SERC GPU **Precog student, SERC GPU steward, faculty (informed)**	Assumed impossible; only possible via a professor [E].	Documented request to the owning lab's GPU steward; the steward decides; attribution kept; logged; totals reviewed each term.	Is it idle? For how long? Can the owning lab reclaim it?	Requests always refused (ownership defence, LP7) and the refusals go unexamined.	Some idle capacity used; refusals become visible data for the review.
S8 · Dead desk and fridge mess **Student, amenities or desk steward, lab admin, cleaning staff**	The student must personally approach faculty; staff absorb the mess silently [E].	Reported to the steward → logged → routed to the lab admin; cleaning staff send one photo; care steps apply to labelled items.	Who fixes it, and by when? Is the misuse attributable?	Admin capacity is the real bottleneck; the log fills with unresolved faults.	Faults gain a time-to-fix; mess is traced where labelled; recurring faults reach the review.

How each stakeholder is likely to respond to the full bundle

⚑ PREDICTED RESPONSE BASED ON EXISTING EVIDENCE — simulated, not a research finding

FIELD	Newcomer / junior (UG, MS)	Established holder (senior PhD)	GPU manager POC → GPU steward	New desk / amenities steward
LIKELY BENEFIT	Can read availability; has a named first stop.	Real continuity protected explicitly; fewer "is this free?" interruptions.	A rule to point to instead of persuasion.	A recognised role and influence over the floor.
LIKELY CONCERN	Borrowing a senior's desk still feels socially risky.	Losing the desk; bureaucracy; feeling judged.	More requests; arbitrating between deadlines.	Confronting seniors; unpaid time.
BEHAVIOUR CHANGE	Read cards; use the queue; raise issues with a steward.	Date the card weekly; declare exceptions with end dates.	Check the page; apply lapses; broker deadline requests.	Weekly bay walk; log; respond within 2 working days.
NEW WORKLOAD	Very low.	~1 min/week.	Moderate at peak.	≤1 h/week (cap).
INCENTIVE TO COMPLY	Seated faster; less dependence on a broker.	The claim becomes legitimate and defensible.	Less DM chasing; clearer authority.	Recognition and visible faculty backing.
INCENTIVE TO BYPASS	None needed: asking a senior friend still works, and that is fine.	Mark everything Protected; renew without use.	Ignore requests at peak; informal favours.	Avoid conflicts; let the log lapse.
POWER TO RESIST	Low [E: juniors withhold complaints].	High (seniority and social standing).	Medium (controls access already).	Delegated and fragile.
LIKELY WORKAROUND	Keeps borrowing only through personal ties.	Ritual renewal; leaving belongings as before.	Grants priority to friends quietly.	Escalates every case to faculty.
UNANSWERED QUESTION	Will they actually sit at a stale-card desk? (Test in S1.)	WEAK EVIDENCE: no data on how holders react when asked to release.	How many hours does a deadline week take?	Will anyone volunteer, and will juniors be accepted by seniors?

How each stakeholder is likely to respond to the full bundle

⚑ PREDICTED RESPONSE BASED ON EXISTING EVIDENCE — simulated, not a research finding

FIELD	Faculty / lab heads (SERC 7, Precog 1)	Lab administrator	Cleaning & service staff	Precog students (cross-lab)
LIKELY BENEFIT	Fewer escalations; a factual picture of issues; evidence for split decisions.	Faults arrive clearly described and routed.	Stop silently absorbing desk-eating mess.	A documented path to idle SERC capacity.
LIKELY CONCERN	Loss of control; cross-lab politics; time.	More work for a handful of staff [E].	Being blamed, or asked to police students.	Refusal; being treated as guests.
BEHAVIOUR CHANGE	Delegate small decisions; back stewards; attend one review.	Accept steward-routed faults; hold bagged belongings for 14 days.	Send one photo to the amenities steward.	Request through the SERC GPU steward.
NEW WORKLOAD	45 min/term + occasional cases.	Small increase.	Small.	Low.
INCENTIVE TO COMPLY	Less firefighting.	Fewer vague complaints.	Less recurring mess.	Access for overnight or idle slots.
INCENTIVE TO BYPASS	Keep deciding case by case.	Deprioritise student-routed faults.	Stay silent as today.	Go through a professor as today.
POWER TO RESIST	Highest: can veto everything.	Medium.	Low: execute, never decide [E].	Low (1 Precog faculty vs 7 SERC).
LIKELY WORKAROUND	Approve the rules but never back a steward publicly.	Pushes everything to the institute ticket desk.	Complain only to their supervisor.	Personal networks.
UNANSWERED QUESTION	Will both labs' faculty agree to a joint review?	Real capacity for repairs within weeks, not months.	Will the supervisor allow the reporting route?	Will SERC stewards approve, and on what grounds?

TEST AGAINST SYSTEM OBJECTIVES · EVALUATION FRAMEWORK

Ten criteria, each with a signal we could actually observe in a 4-week pilot

CRITERION	DEFINITION HERE	QUALITATIVE INDICATOR	QUANTITATIVE INDICATOR	SUCCESS SIGNAL	FAILURE SIGNAL
Effective availability	Physically free desks and GPU hours become usable.	Fewer "is this free?" DMs	Stale desks borrowed per week; unstarted reservations lapsed	Stale desks get used within a week	Cards everywhere, desks still empty
Newcomer access	Access without an insider broker.	Newcomer interviews (3–5)	Days from arrival to a first usable desk	Seated within 2 days, no senior needed	Access still runs through seniors
Continuity protection	Legitimate runs, rigs and absences untouched.	Holders feel protected, not exposed	Running jobs or protected setups disrupted (target 0)	Zero wrongful lapses	Any running job or rig disturbed
Coordination cost	People and messages needed per access.	Self-reported effort	People contacted per request	Falls against the baseline week	New steps added to simple cases
Predictability	Comparable cases handled alike.	Can students state the rule?	Share of steward decisions citing a rule	Most decisions cite a rule	Seniority still decides
Flexibility	Exceptions and judgment remain possible.	Exceptions feel reachable	Exceptions requested vs approved; share of desks Protected	Exceptions approved, and under 30% of desks Protected	Refused by rule, or exception inflation
Burden	Load on stewards, faculty, admin and staff.	Complaints of overload	Steward hours per week (cap 1)	Within the cap	Over 2 h for 2 weeks running
Trust and cooperation	Lending and the ask-first norm survive.	People still lend informally	Informal lending mentioned in interviews	Lending continues	"Not my job, the rule says"
System learning	Issues reach people who can change rules.	Stewards and faculty read the log	Log entries; issues discussed; rules changed	≥1 recurring issue reaches the review	Log empty or unread
Gaming resistance	Strategic behaviour stays low.	Stewards notice patterns	Ritual renewals; false deadlines; dummy jobs	Rare and caught early	Ritual becomes the norm

What improves, what gets worse, and who pays

CRITERION	1 · Claim Signals	2 · Default-Decay Claims	3 · Proportionate Care
WHAT IMPROVES	Readability; newcomer confidence; stale claims become visible.	Dead claims recovered; neutral rules replace confrontation; GPU idle-holds released.	Shared access kept; staff get a reporting role; fairness of correction.
WHAT DOES NOT CHANGE	Who holds what; ownership; GPU quota.	Faculty and ownership decisions.	Desks and GPUs; the Saturday clear.
WHAT GETS WORSE	Small upkeep for holders; Protected may inflate.	Feels like control if introduced badly; risk of false lapse.	Slower than a blanket rule; needs someone to send reminders.
WHO BENEFITS · WHO PAYS	Newcomers and juniors benefit; holders pay a small cost.	Waiting users benefit; long-term holders and whoever confirms lapses pay.	All amenity users and cleaning staff benefit; the reminder-sender pays.
POWER SHIFT	Newcomers gain information; seniors lose the monopoly on occupancy knowledge.	Rules gain; incumbents lose default permanence.	Faculty keep the blanket-restriction decision but must justify it in writing.
LOOPS WEAKENED · NEW LOOP	Weakens R2 (information asymmetry). New loop: Protected inflation → signals lose meaning.	Directly weakens R1 (the desk spiral) and R1/R3 hoarding. New loop: ritual renewal.	Weakens B1 (misuse → blanket restriction).
PEAK DEMAND · IF IGNORED	Helps at peak (it shows expected release); if ignored, cards go stale and nothing changes.	Strongest effect at peak; if ignored, unenforced rules erode trust in all rules.	Not demand-sensitive; if ignored, drift back to blanket rules after the next incident.
AFTER 3–6 MONTHS · DEMAND GROWTH	Without a rule behind it, 3–6 months of decay back into jackets and names.	Renewal rituals likely emerge by 3–6 months unless someone talks to holders.	Works only if labelling becomes habit within the first month.
WHAT THE SCENARIO WALKTHROUGH REVEALED	S1 and S3: Protected needs an end date or it becomes a permanent claim.	S3, S5, S6: immediate reassignment hurts travellers; lapse must be tiered.	S8: staff must report, never enforce; correction must be private.
STILL UNTESTED	Whether people update weekly.	Holder reactions (no evidence yet); the right day counts.	Anonymous misuse; whether faculty reopen coffee.

What improves, what gets worse, and who pays

CRITERION	4 · Stewards + Ladder	5 · Allocation Review
WHAT IMPROVES	A first stop exists; exceptions decided by people; issue log created.	Allocation tracks headcount; recurring issues become rules.
WHAT DOES NOT CHANGE	Faculty authority over allocation.	Day-to-day access.
WHAT GETS WORSE	The heaviest new burden, on unpaid students.	Faculty time; cross-lab tension surfaces.
WHO BENEFITS · WHO PAYS	Juniors benefit most; stewards pay; faculty gain time.	The whole floor benefits long-term; faculty pay in time.
POWER SHIFT	Stewards gain limited authority; the informal senior broker loses some.	Allocation moves from history to evidence.
LOOPS WEAKENED · NEW LOOP	Weakens B2 (conflict avoidance) and the broken feedback link. New loop: steward as gatekeeper.	Closes the missing learning loop; addresses ossification. New loop: review ratifies the status quo.
PEAK DEMAND · IF IGNORED	Overload risk at deadlines; if ignored, cases bounce to faculty as today.	No direct effect at peak.
AFTER 3–6 MONTHS · DEMAND GROWTH	Burnout at term end without a cap, pairing and recognition.	Value appears only after 1–2 terms; demand growth makes it more valuable.
WHAT THE SCENARIO WALKTHROUGH REVEALED	S2 and S4: peak load exceeds 1 h; a junior steward alone lacks standing.	S7: cross-lab decisions sit better with the owning steward than a committee.
STILL UNTESTED	Volunteer supply; legitimacy over time.	Faculty attendance; cross-lab politics.

From intervention to unintended consequence

Each chain runs intervention → intended response → stakeholder adaptation → secondary effect → unintended consequence.

INTERVENTION	INTENDED RESPONSE	STAKEHOLDER ADAPTATION	SECONDARY EFFECT	UNINTENDED CONSEQUENCE	LIKELIHOOD	EVIDENCE FOR THE RATING
1 · Signals	Holders tick a state	Tick Protected "to be safe"	Protected becomes the default	**Signals lose meaning; newcomers are back to guessing**	**LIKELY**	Belongings already act as enduring claims [E]
1 · Signals	Expected release shown on the GPU page	Holders feel watched	Fields left blank or gamed	**Surveillance effect; less honest data**	**POSSIBLE**	No direct evidence; follows from the privacy tension
2 · Decay	Claims lapse unless renewed	Renew every week without use	Claims persist on paper	**Ritual renewal; the R1 spiral continues**	**LIKELY**	Claims persist today with no rule at all [E]
2 · Decay	Unstarted reservations lapse after 2 h	Start dummy jobs to hold the slot	Compute wasted	**Availability gets worse than today**	**POSSIBLE**	Idle-but-held slots already exist [E]
2 + 4 · Priority	Dated deadlines get priority	Everyone declares a deadline	The GPU steward is flooded	**False urgency; the rule is discredited**	**LIKELY at peak**	Urgency persuasion is already the norm [E]

From intervention to unintended consequence

Each chain runs intervention → intended response → stakeholder adaptation → secondary effect → unintended consequence.

INTERVENTION	INTENDED RESPONSE	STAKEHOLDER ADAPTATION	SECONDARY EFFECT	UNINTENDED CONSEQUENCE	LIKELIHOOD	EVIDENCE FOR THE RATING
3 · Care	Named correction for labelled items	Stop labelling food	Misuse becomes anonymous	**Pressure returns for a blanket rule**	**POSSIBLE**	Pattern of blanket responses [E]
4 · Stewards	Stewards arbitrate	Juniors defer; seniors lobby stewards	Steward becomes a gatekeeper	**A new informal hierarchy**	**POSSIBLE**	Seniors broker access today [E]
4 · Stewards	Unpaid role with duties	Volunteers drift after a few weeks	Log stops; cases go back to WhatsApp	**Burden shifting, then collapse**	**LIKELY without a cap**	Complaints already die in WhatsApp [E]
5 · Review	Joint faculty review	Each lab defends its share	Review ratifies the status quo	**Formality; LP6/LP7 unchanged**	**POSSIBLE**	Ownership boundary is strong [E]; faculty views untested
All	More written rules	Newcomers face more to learn	Onboarding burden	**Rules feel bureaucratic**	**POSSIBLE**	Newcomers learn norms informally [E]

LIKELY = the adaptation already happens under today's arrangement [E]. POSSIBLE = plausible from the mechanism, no direct evidence. SPECULATIVE = none rated this way; risks without any basis were dropped.

Refining the Interventions

No intervention is dropped because a risk exists. Each gets the smallest change that reduces its main failure mode.

INTERVENTION	VERSION 1	MAIN RISK	WHAT THE WALKTHROUGH REVEALED	MINIMUM REFINEMENT → VERSION 2	TRADE-OFF REMAINING	STILL TO TEST
1 · Signals	Four states on every desk card	Protected inflation; stale cards	S1, S3: Protected with no end date = a permanent claim	**Protected needs an end date + steward initials; "last here" date line**	One extra step for real continuity	Do holders date the card?
2 · Decay	Fixed N days, then reassign	Punitive feel; false lapse; ritual renewal	S3, S5, S6: travellers hurt; renewal ≠ use	**Tiered: 7 days borrowable → 21 days message → 5 days reply; running jobs never lapse; repeat renewal → conversation**	Dead desks recovered more slowly	Are 7 / 21 / 5 the right numbers?
3 · Care	Named correction for misuse	Anonymisation; shaming	S8: staff must report, not enforce	**Private word, never public; blanket restriction needs a written reason and a review date; run by the amenities steward**	Slower than a blanket rule	Faculty agreement
4 · Stewards	One steward per resource, broad rights	Bottleneck; junior vs senior; burnout	S2, S4: peak load exceeds 1 h; a junior alone lacks standing	**Cross-lab pairs mixing seniority; 1 h/week cap with overflow to faculty; decide only by written rule; faculty back the role publicly; GPU steward = existing POC**	Two people must coordinate	Volunteer supply; time at peak
5 · Review	Termly joint review decides allocations	Formality; faculty time; politics	S7: cross-lab calls fit the owning steward better than a committee	**45-min fixed agenda; decisions only from log counts; cross-lab decided by the owning steward; first run as a mock at pilot end**	Structural change is slower	Faculty attendance

WHAT CHANGES, LAYER BY LAYER

Nine layers of change, each kept to the minimum

INFORMATION

Claim state on every desk; expected release and OK-to-pause on the GPU page; a one-line issue log.

RULES

7 / 21 / 5-day desk rule; locker renewal each term; 2-h GPU lapse; continuity exceptions; care steps.

ROLES

Desk, amenities and GPU stewards in cross-lab pairs, one term each; cleaning staff may report (not enforce).

PROCESS

Peer → steward (≤2 working days) → faculty (≤1 week); request → time-boxed use → confirmed release.

RESPONSIBILITIES

Holders date their card weekly and declare exceptions. Stewards walk the bay weekly and log. Admin fixes routed faults.

TOOLS

Paper cards in sleeves; two fields on the existing student-built GPU page; a shared sheet for the log. Nothing new to install.

PHYSICAL

Card sleeves at desk edges; a bay notice naming the stewards; a label-and-date station at the fridge.

GOVERNANCE

Decision-rights matrix; blanket restrictions need a written reason and a review date; one joint termly review.

FEEDBACK

Log → termly review → change a number, a state or a role → post the change on the bay notice.

WHAT DOES NOT CHANGE

The useful parts of today's system are kept on purpose

Phase 2 found courtesy and cooperative borrowing working for established members [E]. The design extends that to everyone instead of replacing it.

ASK-FIRST NORM AND INFORMAL LENDING

Still the first step for every request, and still enough in most cases.

FACULTY AUTHORITY

Over ownership, the lab desk split, restrictions and anything contested.

ATTRIBUTION OF PURCHASED EQUIPMENT

Who paid stays visible; cross-lab use is by request and revocable.

GPU QUOTA TIERS AND THE 6-H INTERACTIVE CAP

Untouched; the new fields sit on top of them.

THE SATURDAY FRIDGE CLEAR

Kept as the weekly reset; labelling adds attribution.

MEETING-ROOM BOOKING

Already formal and working [E]; out of scope.

SENIORS MENTORING NEWCOMERS

Encouraged, not required; the bay card removes the dependency, not the relationship.

CASE-BY-CASE JUDGMENT

Every exception and every contested case is decided by a person.

USER JOURNEY · OPERATING FLOW

How the system runs for its two most common journeys

JOURNEY A · A NEWCOMER NEEDS A DESK

ARRIVE
Bay notice names the desk steward; 10-min norms briefing.

→

READ CARDS
FREE or stale (7+ days) → sit for the day. Protected → never.

→

ASK FIRST
Courtesy stays: a quick word with neighbours or the holder.

→

JOIN QUEUE
The steward adds them; stale claims are chased (21 + 5 days).

→

OWN DESK
Offered in queue order; dated card from day one.

JOURNEY B · A STUDENT NEEDS A GPU BEFORE A DEADLINE

CHECK PAGE
Free slots, expected releases and OK-to-pause flags are visible.

→

RESERVE
Within the quota; expected release required; start within 2 h.

→

DEADLINE ≤72 H?
Request priority → GPU steward asks "Yes" holders first.

→

CONSENT
A holder agrees to pause; nothing is stopped without consent.

→

RELEASE + LOG
Slot freed on the page; one log line if the steward was involved.

WHEN SOMETHING GOES WRONG

Any journey: peer first → steward within 2 working days by written rule → faculty within 1 week if contested. Faults and recurring mess go through the steward to the lab admin. Every step leaves one line in the log.

REVIEW LOOP · HOW THE SYSTEM LEARNS

Use → observe → review → modify → publish

This is the feedback loop Phase 2 found missing: demand that disappears must stop looking the same as demand that was met.

USE

Cards, page fields and the ladder run day to day.

→

OBSERVE

Weekly bay walk; one log line per case; steward hours.

→

REVIEW

Stewards check in mid-term; the joint review sits at the end of each term.

→

MODIFY

Change a number (7/21/5, 2 h), a card state, a role or the desk split.

→

PUBLISH

Post the change on the bay notice and GPU page; it applies next term.



the loop repeats every term

HOW A RECURRING ISSUE BECOMES A POLICY QUESTION

An issue logged 3+ times in a term is automatically on the review agenda. The review can change a rule, a role or the allocation, and records the decision on the bay notice.

WHAT THE REVIEW CAN AND CANNOT DO

Can: numbers, states, steward roles, desk split, cross-lab guidance. Cannot: override ownership of purchased equipment or the institute's rules.

Recommended: a staged minimum bundle

Make claims readable, let them decay by default, and give each resource a named first stop. Then let one short review turn what the log shows into rule and allocation changes.

STAGE 1 · CORE SYSTEM

1 · Claim Signals

Dated desk cards; two fields on the existing GPU page

2 · Default-Decay

Tiered 7 / 21 / 5-day desk rule; 2-h GPU lapse; continuity exceptions

4 · Stewards

Cross-lab pairs, capped at 1 h/week; GPU steward = existing POC; issue log

3 · Care → inside 4

Run by the amenities steward: label → reminder → private word → individual limit

STAGE 2 · EVERY TERM

5 · Review (light)

One 45-min joint SERC–Precog review each term, using the issue log

NOT RECOMMENDED NOW

A new booking app or dashboard · automated idle detection or sensors · algorithmic allocation · pooling GPUs across labs beyond the request path · a coffee rule change (sent to faculty as a separate proposal)

WHY THIS SET: it covers the Phase 3 chain LP3 → LP4 → LP5 with the fewest parts; 2 cannot work without 1 and 4; 3 costs little inside 4; 5 is the only way the system learns. Every part can be removed without leaving damage behind.